

Section 95 — Geometric Optimization And Delaunay Structure

A few boundary witnesses determine the optimum

This lesson derives A–C. Problem D is editorial-only.

A. Randomized minimum enclosing circle

Shuffle the points. Maintain the minimum circle for the processed prefix. If a new point lies outside, it must lie on the new boundary. Rebuild with that point fixed; whenever an earlier point is outside, both become boundary points; a third violation fixes the unique circumcircle.

The nested loops look cubic, but random insertion order gives linear expected time: a circle determined by at most three boundary points changes with probability at most $3/i$ at step i . The correctness proof is deterministic; only the running-time analysis uses randomness.

B. Incremental Delaunay triangulation

A triangle is Delaunay when its open circumdisk contains no input point. Begin with a supertriangle. For each point, remove every triangle whose circumcircle contains it. Those bad triangles form a cavity; edges occurring once on their boundaries form its polygon. Join the new point to every cavity boundary edge.

The in-circle determinant must be interpreted with the triangle orientation. With counterclockwise triangles, a positive sign means inside. Generic-position constraints remove zero-sign ambiguity; production implementations still need a symbolic or deterministic tie policy.

C. Euclidean MST from Delaunay edges

Every Euclidean MST edge belongs to the Delaunay triangulation. One proof uses the empty diameter disk: if an MST edge had a point strictly inside that disk, replacing it by a shorter edge across the induced cut would contradict minimality. Therefore construct Delaunay edges, sort only those by length, and run Kruskal.

Recognition guide

When an optimum circle or proximity edge seems to depend on all points, ask which small witness set makes it tight. Enclosing circles need at most three boundary points; Voronoi vertices are fixed by three equidistant sites; MST edges admit an empty-region certificate.

Why the enclosing-circle rebuild is complete

If point p lies outside the optimum circle for the processed prefix, every new optimum must have p on its boundary; otherwise the circle could move or shrink slightly while still covering the old points and p . With p fixed, an outside point q similarly forces the diameter circle through p, q or a later circle with both on its boundary. A third violating point fixes their circumcircle. Thus the nested loops enumerate all possible support-set sizes, not arbitrary repair heuristics.

Cavity and empty-region invariants

In incremental Delaunay, the removed triangles are exactly those whose circumdisks conflict with the inserted point. An edge appearing twice is internal to the cavity; an edge appearing once lies on its boundary. Connecting the new point to boundary edges fills the cavity and restores the empty-circle condition locally.

This clean description assumes generic position. Decide in advance how to handle duplicate points, collinearity, and cocircular points. Random insertion does not fix an inconsistent in-circle predicate. Prefer an exact integer determinant when coordinate bounds permit; otherwise document the tie rule.

Verification strategy

For minimum enclosing circle, check containment and compare with all circles defined by one, two, or three points on small sets. For triangulation, verify edge incidence and the empty-circumcircle condition. For Euclidean MST, compare the Delaunay-edge Kruskal weight with complete-graph Kruskal on small random sets, including grids and nearly collinear configurations.

Exercises

A–C develop enclosing disks, empty-circle triangulation, and proximity sparsification. D asks you to extract the largest finite empty circumcircle.